

LAHORE COLLEGE FOR WOMEN UNIVERSITY, LAHORE
Mid / Final Examination 2025

Class: BS-CS

Paper No: CS-425

Note: Attempts all questions.

Subject: Information Security

No. of credit hours: 3

Semester: V

Max Marks: 50

Term: Final

Time allowed: 3hrs

S.No	Questions	M	BT	CO	PO
Part -A(Short Answer Questions)					
1	What is meant by security attack? Explain various types of security attacks.	2	L1	CO1	PO1
2	Why do we need Diffie Hellman algorithm?	2	L2	CO2	PO3
3	Convert the Given Text "CRYPTOGRAPHY" into cipher text using Rail fence Technique. <i>k=3</i>	2	L3	CO1	PO1
4	What is the difference between a MAC and a one-way hash function	2	L2	CO2	PO3
5	Describe three ethical issues that should be considered when using computers.	2	L3	CO2	PO3
Part- B(Long Answer Questions)					
6	Use the Chinese Remainder Theorem to find the value of X that simultaneously solve following :- $2X \equiv 6 \pmod{14}$ $3X \equiv 9 \pmod{15}$ $5X \equiv 20 \pmod{60}$	5	L3	CO4	PO3
7	Consider a Diffie-Hellman scheme with prime $n = 11$ and $a = 7$, then a. If user A pick his private key $X_A = 6$, what is A's public key Y_A ? b. If user B pick his private key $X_B = 9$, what is the secret key K shared with A?	5	L3	CO4	PO4
8	Differentiate Block cipher and Stream Cipher . Draw general depiction of DES Encryption Algorithm.	5	L4	CO2	PO3
9	Name the protocols that provide security in IPSec,	5	L2	CO4	PO5

***BloomsTaxonomyLevel(BT)**

(L1-Remembering;L2-Understanding;L3-Applying;L4-Analyzing;L5-Evaluating;L6-Creating)

Course Outcomes (CO), Program Outcomes (PO)

Flame

27-11-2025